

11. Intellectual property infringement: a persistent threat to the world economy

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1. INTRODUCTION

Intellectual property (IP) constitutes a significant share of many companies' equity. It is, in most cases, the result of extensive investments in research and development, careful brand management, and a consistent pledge to high quality, reliability, and exclusiveness. In addition, the complex nature of the intellectual property landscape makes modern, innovative industries prone to rely simultaneously on multiple types of intellectual property in their operations.

However, the growing momentum of globalization – including the growing role of economies where IP is more difficult to protect – plus a general trend in favor of dismantling border controls to ease the flow of international trade, and the increasing integration and interaction between organizations in disparate locations, require new measures to protect these assets and safeguard companies from unfair competition.

The intangible nature of IP makes infringement easy, and in today's globalized and interconnected world, illicit trade in IP-infringing goods becomes a vital threat to modern, innovation-driven economies. Bad actors that drive illicit trade in IP-infringing goods exploit governance gaps and abuse trade facilitation solutions.

Reliable statistical evidence on the scale and magnitude of IP infringement is limited. The best existing information is drawn from data on customs seizures that provide only a partial picture of the situation, as the operations are targeted and focus only on selected types of intellectual property – mostly trademarks. However, this quantitative picture, together with complementary qualitative information, supports the claim that illicit trade in IP-infringing goods is global and destructive.

This chapter addresses the risks associated with intellectual property infringements, including patent infringements, within the context of global trade. It contributes to the discourse by providing evidence and offering a

comprehensive, quantitative overview of the challenges posed by the illicit trade of goods that violate IP rights. These goods, often smuggled, exploit governance voids and manipulate contemporary trade mechanisms, with perpetrators adopting hit-and-run tactics. The magnitude of this risk is substantial, highlighting the pervasive threat of IP infringement across international markets.

Furthermore, the discussion emphasizes the necessity for companies to consider the potential risks of infringement when adopting modern IP solutions. This chapter thus underscores the importance of a proactive approach to IP management, acknowledging the significant challenges that IP infringements pose to the integrity of global trade and the innovation ecosystem.

This chapter is structured as follows. The next section discusses the existing literature. The following section discusses the great IP-dependency, including patent dependency, of modern industries that rely on IP bundles. The third section highlights the current socio-economic landscape in the context of IP infringements, focusing on two key enabling factors: globalization of supply chains and the persistence of enforcement gaps. The fourth part presents extracts from the existing quantitative and qualitative evidence on illicit trade in IP-infringing goods. The last part concludes.

2. LITERATURE OVERVIEW

Illicit trade in goods that infringe intellectual property is a relatively new phenomenon that escapes traditional academic taxonomies. Consequently, literature illustrating the threat of IP infringement, including patent infringement, in the context of globalization and world trade is rather limited. Most information on trade in IP-infringing goods comes from the press and trade magazines such as *The Economist*,¹ or thematic reports published by think tanks or industry associations.²

As for the academic discourse on IP infringement in the context of globalization, it has been evolving since its emergence in scholarly discussions in the late 1970s. Until today, research on IP infringement has been fragmented across various academic disciplines such as strategic management, marketing, and logistics, without forming a distinct field of study. Studies emphasize the long-standing issue of physical goods counterfeiting, acknowledging the difficulty in measuring the scale of this illicit trade due to its illegal nature (Staake et al., 2009). Efforts by organizations like the U.S. General Accountability Office highlight the challenge of quantifying the counterfeit market, underscoring the need for better data and strategies to combat this problem (GAO, 2010).

Research also delves into the smuggling of IP-infringing goods, indicating persistent challenges despite efforts to curb such activities. The literature

connects IP infringement with broader illicit trade forms, from counterfeit goods to environmental crime, underscoring its negative impacts on human development and the Sustainable Development Goals. Proposed solutions include improving data collection, integrating anti-IP infringement measures into global trade frameworks, and targeting problem states facilitating illicit trade (Chaudhry and Zimmerman, 2013; Lallerstedt, 2022; Gwardzińska and Chackiewicz, 2022).

Regarding the supply and demand of IP-infringing products, China is identified as the primary source, with international attention focused on various countries' IP environments (EUIPO-Europol, 2022; WCO, 2022). Consumer demand for IP-infringing goods, and counterfeit goods in particular, driven by factors explored in historical and marketing perspectives, suggests avenues for further research (Shelley, 2012; Chaudhry and Zimmerman, 2013b; Latif et al., 2018). The rise of e-commerce has exacerbated the problem, offering counterfeiters new avenues to penetrate legitimate markets, thereby necessitating a multifaceted approach to IP protection that includes enhancing legal frameworks and international cooperation (Shelley, 2012; Kennedy, 2020).

In summary, the literature on IP infringement spans various research areas, highlighting the complexity of combating counterfeit goods in a globalized and digitalized world. The challenge of measuring illicit trade, the diverse sources and drivers of counterfeit products, and the evolving strategies for IP protection underscore the need for interdisciplinary research and international collaboration to address these issues effectively.

3. IP INTENSITY OF MODERN INDUSTRIES

Innovative industries in modern economies increasingly rely on intellectual property. Consequently, quantitative studies highlight that for many innovative countries, intellectual property brings significant contributions to IP-intensive industries in terms of value added, wages, and employment.

Rather than using only one type of intellectual property, firms tend to rely on several types of IP in their activities (for example, patents and trademarks). Such IP-intensity is desirable from a social perspective because intellectual property, and – more broadly – innovative processes protected by IP, spur economic growth and enhance the welfare of nations. However, given the intangible and non-rival nature of intellectual property, heavy reliance on IP makes innovative firms vulnerable to infringement and appropriation of the key components of their business models.

The IP-intensity of modern industries has been analyzed for several economies for which data are available. Conclusions were consistent across analyzed countries. First, modern, innovative industries tend to rely on intellectual property to a growing extent. Secondly, most industries tend to use several

types of intellectual property simultaneously – such as trademarks, patents, and copyrights – and industries that only use one intellectual property right are not common (EUIPO-EPO, 2022; USPTO, 2022; UKIPO, 2020).

Regarding patents, the procedure to determine the degree of patent intensity of an industry is rather straightforward. The following example comes from an EU-focused study (EUIPO-EPO, 2022), which uses a three-step procedure:

- In the first step, the total number of patents protected under the European Patent Convention (EPC) is counted. This value is then reported as the *absolute patent intensity*.
- In the second step, the *absolute patent intensity* is adjusted at the industry level by the volume of employees in that industry. The result is called *relative patent intensity*, and it is reported for each industry.
- In the final step, for all industries that rely on patents, the aggregate, *average relative patent intensity* is calculated, using the volumes of employment as a weight. Patent-intensive industries are those with their *relative patent intensities* higher than the *average relative patent intensity*.

A similar procedure has been carried out for trademarks to determine trademark-intensive industries.

The detailed scoping of IP-intensive industries highlights that most of them employ, to a high degree, more than one type of intellectual property rights. For example, “Manufacturing of ceramic insulators and insulating fittings” is an intensive user of trademarks and patents, while “Manufacturing of other transport equipment n.e.c.”³ intensively uses trademarks, designs, and patents (EUIPO-EPO, 2022).

The overlaps in terms of intensity of types of IP used are more the rule than the exception. In fact, as demonstrated by a recent economic analysis presented in the quoted EUIPO-EPO study, most industries have used a wide bundle of intellectual property rights. Moreover, trademarks are used intensively by most of the industries (77%). What is relevant is that 79% of industries that use trademarks also use other IP rights, mostly patents, copyrights, and designs. Importantly, the trademark-patent bundle tends to be used intensively in the manufacturing sector.

Conversely, few industries used only one intellectual property right. There were only 23 patent-only intensive industries (6.4%). In addition, the 23 industries that are exclusively patent intensive are mostly business-to-business, gathering intermediate suppliers that offer their products and services to other industries, not to end consumers. These industries include, for example, “casting of steel” and “engineering activities and related technical consultancy”.

Consequently, branding of their products and services is of lesser importance, as they rely on direct contact with the demanders.

Two takeaways from this analysis of IP-intensity of modern industries are relevant in the context of this chapter:

- First, in modern, innovative economies, a large volume – more than three-quarters – of the industry relies on intellectual property. IP is by nature intangible, non-exclusive, and non-rival. This makes it a lucrative target for criminals that seek large and easy rents.
- Second, most IP-users employ a bundle of intellectual property rights simultaneously. Only a small minority of industries rely on one type of IP only. Patent-trademark is the most intensively used bundle in the manufacturing sector.

4. GLOBALIZATION AND CONDITIONS FOR IP INFRINGEMENTS

The enhanced IP intensity of modern economies, together with progressing trade facilitation, brings welfare and enhances economic growth. The logistical and technological solutions facilitate trade and allow a more efficient allocation of industrial activity. However, at the same time, these phenomena unwittingly create suitable conditions for IP breaches to emerge. Specifically, there are two interdependent factors that determine the rising volume of infringement of intellectual property rights in the context of world trade.

First, the growing complexity of supply chains, and their increasing geographical span, make them prone to infiltration by bad actors that seek to steal the intellectual property of legitimate rights owners. This can happen particularly in jurisdictions with good industrial and logistical infrastructure, but, at the same time, with limited respect for IP, low transparency, and insufficient governance standards.

Second, even in countries with sound governance frameworks and high-transparency institutions, those in charge of IP enforcement and countering infringements are often not adequately empowered to deal with this complex challenge (OECD, 2018). These institutions are often facing numerous competing priorities and are not adequately empowered to deal with the specific characteristics of illicit trade in IP-infringing goods. Criminals who run illicit trade networks are well aware of these factors. They design their operations not only to maximize their profits but also to minimize the risk of detection by using sophisticated techniques and abusing modern logistical solutions.

4.1 Complex Supply Chains and Soft Spots

The global supply chains that distribute goods worldwide are intricate and multifaceted, making them vulnerable to infiltration by criminals who introduce counterfeit products into these chains. These supply chains involve numerous intermediate players and span across countries with varying judicial systems and levels of intellectual property enforcement. While some economies may have robust production and logistics infrastructure, they may lack adequate IP protection, creating opportunities for IP infringement.

Criminals who rely on the breaching of IP rights in their operations carefully consider several factors when engaging in the illicit production and distribution of infringing goods. They take into account the location and cost of production while minimizing the risk of interception and prosecution. The decisions made by bad actors are influenced by factors such as the potential profitability and size of the markets for the counterfeit products, technological and logistical considerations related to production and distribution, and the risk and consequences associated with detection by law enforcement agencies (EUIPO-Europol, 2022).

In terms of supply of IP-infringing goods, criminals often ship them through several transit points. These transit points serve multiple purposes for the criminals. Firstly, they facilitate the falsification of documents to obscure the original point of departure of the counterfeit goods. Secondly, these transit points establish distribution centers for counterfeit products, allowing for their efficient dissemination. Thirdly, IP infringers utilize these points to repackage or re-label the goods, further obscuring their origins. Importantly, while local enforcement authorities often target counterfeit goods at their destination, goods in transit are frequently overlooked, making them less likely to be intercepted (OECD/EUIPO, 2017).

The existing quantitative evidence indicates that corruption levels and the quality of intellectual property protection are crucial drivers of illicit trade in IP-infringing goods. Jurisdictions with widespread corruption and poor control over it tend to have a higher share of illicit products in their exports. Conversely, jurisdictions with strong intellectual property protection measures experience a lower share of such exports (OECD/EUIPO, 2018).

Poor governance and low respect for intellectual property rights are not the sole conditions for IP infringement. However, in economies with weak governance and poor respect for IP, the enabling impact of other factors, such as efficient logistics and transportation systems, is amplified (OECD, 2018, OECD/EUIPO, 2018 and 2018).

From a macroeconomic perspective, the propensity of a country to export counterfeit goods is influenced by logistical and industrial factors as well as governance factors. Countries with a high share of manufacturing in their

economy, low labor costs, and suitable labor market regulations are more likely to have favorable conditions for producing goods, including IP-infringing ones. Additionally, factors such as logistical performance, the presence of Free Trade Zones (FTZs), efficient customs clearance processes, competitively priced shipments, and high-quality transport infrastructure contribute to the export and transit of all goods, including IP-infringing ones.

The findings outlined above highlight the effectiveness of legal and regulatory frameworks, particularly in terms of IP protection, in combating IP infringement in the trade context. These frameworks provide rights holders with the means to take legal action against infringing parties and seek compensation for their losses. Strong frameworks act as deterrents to illicit activities, while weak frameworks are viewed as permissive, encouraging counterfeiting. However, the impact of laws and regulations depends on their enforcement. Insufficient resources allocated to enforcement or a lack of enforcement by public authorities diminish the value of these legal measures.

Corruption plays a significant role in undermining enforcement efforts. It allows illicit production facilities to operate undetected, permits the mixing of fake and genuine goods within distribution channels, and leads to the shelving of cases and lack of action on complaints. In such circumstances, even stringent laws may fail to deter counterfeiting and piracy.

4.2 Persistent Enforcement Challenges

Even in countries with sound governance frameworks, effective IP enforcement in the trade context poses significant challenges. Criminals who run illicit trade networks employ a range of clever techniques to avoid detection and successfully ship illicit, IP-infringing products across borders. These illegal techniques are tailored to suit the specific characteristics and value of the illicit goods involved. Together with massive volumes of modern trade and competing enforcement priorities, they can significantly reduce the efficiency of enforcement measures.

One commonly used method involves document falsification. For instance, in December 2019, a smuggling operation that transported counterfeit products from China through the ports of New York and New Jersey was intercepted. This operation included 22 containers filled with counterfeit sneakers, which could have fetched a staggering USD 472 million if they were genuine. To deceive authorities, the ship manifests contained false information, describing the merchandise as ventilation fans, vases, and plastic hangers. Moreover, the container importers used the identities of legitimate companies on customs forms, but provided contact details belonging to the counterfeiters, who used disposable phones and email accounts obtained through false identities to conceal their activities. Once the containers passed through customs, they

were shipped to self-storage facilities, where the counterfeit items were dismantled for sale and distribution to wholesalers and retailers. The investigation revealed that 107 other container shipments were connected to the counterfeit importers, suggesting a substantial volume of undetected counterfeit trade across the U.S. border (OECD/EUIPO, 2021a).

Another technique employed by counterfeiters involves physically manipulating the products to deceive detection. For instance, in 2018, U.S. authorities dismantled a counterfeiting ring based in New York that smuggled nearly 400,000 pairs of counterfeit Air Jordans into the country, potentially causing Nike a loss of over USD 70 million in revenue. The counterfeit sneakers were initially manufactured without any identifying marks, and the fake logos were added once the shoes cleared customs. Similarly, in October 2019, a person was arrested for shipping fake Timberland and Ugg boots worth over USD 5 million from China to the New York area. In the case of the Timberland footwear, counterfeiters attempted to avoid detection by attaching a shoe insert over a fake Timberland logo on the bottom of the boots (OECD/EUIPO, 2021a).

Furthermore, smugglers sometimes resort to concealing illicit goods within larger shipments of legitimate items to avoid detection. This technique is not exclusive to counterfeiters. For example, in July 2020, Italian police seized 14 tons of the drug Captagon, manufactured by the terrorist group Daesh in Syria. The estimated value of the seizure was USD 1.1 billion, making it one of the largest drug busts in the world. Around 84 million tablets, enough to supply the entire European market, were hidden within industrial goods inside shipping containers. To uncover the hidden drugs, police had to use chainsaws to open up the industrial rolling stock and metal gearwheels in which the pills were concealed.

By employing such deceptive methods, smugglers continue to pose significant challenges to authorities in their efforts to combat the circulation of illegal, IP-infringing goods (OECD/EUIPO, 2021a).

5. QUANTITATIVE EVIDENCE

This section presents quantitative evidence on illicit trade in IP-infringing goods. The quantitative analysis primarily addresses trademark infringement, yet its implications extend well beyond this scope. As discussed in the previous sections, companies typically employ bundles of intellectual property protections, meaning that a product subject to seizure often enjoys protection from multiple IP rights. Consequently, data pertaining to trademark infringements can offer insights applicable to other IP categories. This broad applicability is further justified by the observed correlation between trademark and patent intensities, suggesting that industries known for their trademark reliance often also exhibit significant patent activity. Moreover, trademark-intensive

sectors cover a wider array of industries than those primarily focused on patents, and they frequently encompass aspects of patent infringement. Therefore, trademark infringement data not only shed light on the dynamics of trademark-intensive industries but also offer a window into the practices within patent-intensive sectors.

Furthermore, the process of enforcing trademarks in the case of product infringement is comparatively simpler than enforcing patents. The seizure of goods infringing on trademarks involves a standardized procedure that can be executed with the assistance of enforcement agencies such as customs or police. This accessibility to enforcement mechanisms generates a substantial volume of data, which was instrumental in the subsequent analysis. While copyright enforcement also plays a crucial role, particularly in the digital marketplace, the lack of unified data limits the ability to construct a comprehensive overview. Despite this limitation, the analysis benefits from the available information on trademark enforcement, highlighting its ease and the insights it provides into broader IP enforcement challenges and strategies.

5.1 Gauging Illicit Trade in IP-infringing Goods

Measuring illicit trade in intellectual property infringing goods poses several significant data and methodological challenges. Intellectual property infringement is inherently an illegal activity, and obtaining data directly about infringers is not feasible. Therefore, quantitative information on IP infringement often relies on secondary sources such as judicial systems and enforcement agencies.

Currently, the most reliable datasets on IP infringement come from customs agencies and refer to seizures of infringing goods. However, data obtained from the enforcement agencies are typically unstructured and requires extensive manual work to access, clean, and harmonize. Additionally, aggregating this data is difficult due to the diverse legal regimes governing intellectual property worldwide.

Even with these caveats, customs seizure data used in the OECD and EUIPO studies has proven to be the most valuable input, for broader economic analysis. Given the globalization of supply chains and the intangible nature of intellectual property, the risks of IP infringement are global in scope. Therefore, data that describe this phenomenon should allow for a comprehensive global analysis. Aggregated customs seizure data, when properly harmonized, can provide a global overview of the extent and magnitude of IP infringement worldwide.

Customs data represents the foundational element in quantitative analysis. However, to augment their value, additional datasets are necessary to furnish contextual understanding. These encompass general trade statistics, which

serve as a benchmark for analyzing customs seizures data. Furthermore, industry analysis provides essential insights into specific countries of origin and key transit points. Detailed information regarding trade and industrial activities in ports is crucial to evaluate the extent of misuse of particular transit hubs. Finally, a comprehensive array of indicators is imperative to identify potential correlations and address overarching trends related to IP infringement (OECD/EUIPO 2016, 2019 and 2021b).

5.2 Complex Trade Routes, with Few Countries of Origin

Quantitative analysis of illicit trade in IP-infringing goods highlights a great complexity of trade routes with numerous transshipment points.

Of course, the use of transshipment points is a common practice in international trade. These routes serve various purposes and are implemented for several reasons. One reason is that certain ports lack direct connections with specific shipping lines, requiring shipments to be broken down and transported in multiple stages. Another reason for transshipment is the need to change the mode of transport during the journey, such as transitioning from vessel to road transport or from rail to vessel. Additionally, consolidation (combining small shipments into a larger one) and deconsolidation (dividing a large shipment into smaller ones) are factors that contribute to the complexity of trade routes.

IP-infringing products, similarly to legitimate goods, also follow complex trade routes. However, in the case of these goods, complexity is utilized to evade enforcement measures. Criminals involved in the trade of IP-infringing goods constantly adapt their trade routes to avoid detection. These criminals deliberately choose complex routes over simpler ones to minimize the risk of law enforcement intervention. As a result, counterfeit and pirated products continue to flow through intricate trading routes, exploiting intermediary transit points. Criminal organizations play a significant role in this illicit trade, effectively transforming IP infringement into a large-scale illegal production and distribution enterprise with highly complex distribution networks (UNICRI, 2011). Disrupting these networks has proven to be a challenging task.

Efforts to combat the trade in counterfeit goods and dismantle these illicit networks face considerable obstacles due to the inherent complexity and adaptability of the routes used. Developing effective strategies to disrupt these networks and prevent the circulation of counterfeit products remains a significant challenge for law enforcement agencies and regulatory bodies.

Presence and misuse of free trade zones are another significant driver of illicit trade in IP-infringing goods.

The number of FTZs and their relevance work world trade have grown significantly over the past 50 years. In 1970s there was 80 zones in 30 countries with exports totaling USD 6 billion (OECD/EUIPO, 2018b). Fifty years later

the number of FTZs has exceeded 5,000 in 135 countries and keeps growing fast (UNCTAD, 2019).

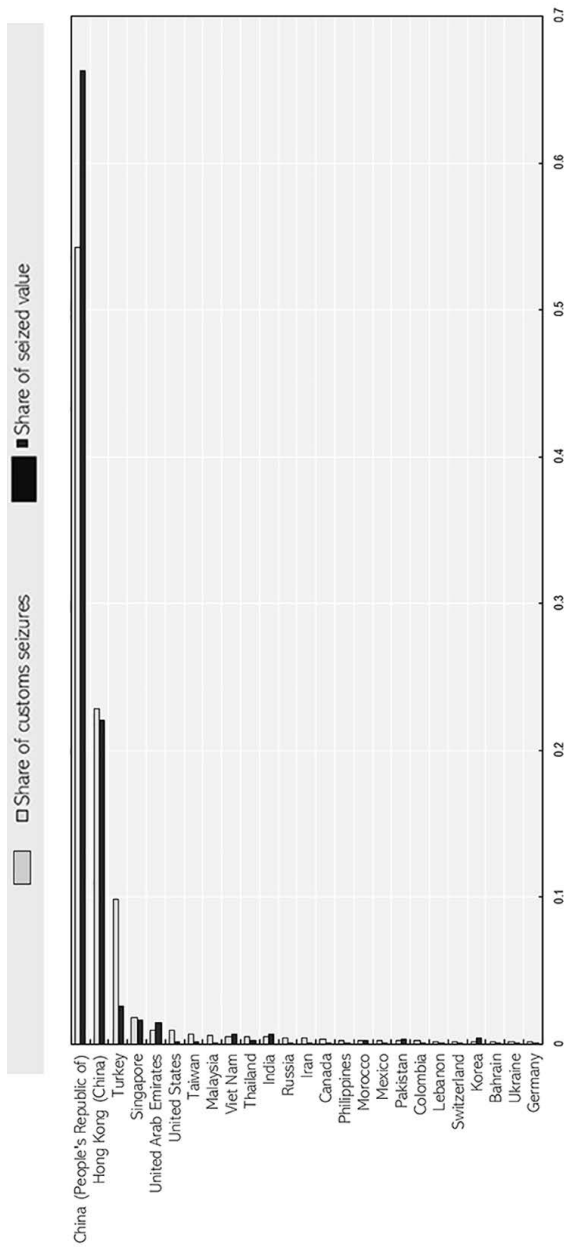
Of course, Free Trade Zones have a significant economic impact, supporting over 68 million direct jobs and generating over USD 500 billion of direct trade-related value added. In addition, over the past several decades, there has been a notable increase in the number of privately owned, developed, and operated Free Trade Zones (FTZs) globally. This rise underscores the private sector's recognition that such facilities can be profitably managed by developers, thereby alleviating the financial burden on government resources. However, these zones' characteristics, which appeal to legitimate traders, also attract organized crime and counterfeiters. Evidence of the misuse of some FTZs by counterfeiters has been mounting. The benefits of transshipment to criminal entities include the ability to obscure the true origin of cargoes, manipulate intellectual property-infringing goods, and operate under minimal regulatory and enforcement supervision. Research indicates that, on average, the proportion of counterfeit and pirated products in exports from economies without FTZs is 50% lower than in those with numerous FTZs (OECD/EUIPO, 2018b).

5.3 “Economies of Provenance”

Existing data also allow for identification of countries of origin of IP-infringing goods. However, distinguishing the origin of seized IP-infringing goods from a transit point is often a challenging task, as data typically do not provide clear information on the original manufacturing point. To address this issue, the term “economies of provenance” has been used, as explained in the OECD-EUIPO (2016) report. Provenance economies encompass both economies where the actual production of infringing goods takes place and economies that serve as transit hubs through which IP-infringing goods pass.

Trade in goods that infringe intellectual property rights can originate from virtually any economy, making the scope of these provenance economies extensive. An analysis of a unified dataset of customs seizures spanning from 2014 to 2016 identified 180 economies of provenance for counterfeit and pirated products. Although the scope is broad, the data also reveal that a limited number of economies dominate the global trade in counterfeit and pirated goods.

Figure 11.1 presents “economies of provenance” in terms of their intensity as indicated by customs seizures. Two indicators can be used for this purpose: the share of seizures from a given economy of provenance in the total number of seizures (irrespective of their size and value), or the share of the total value of seizures from a given economy of provenance in the total value of global seizures. It is evident that Far East Asian countries, particularly China and Hong Kong (China), rank highest in both terms.



Source: OECD/EUIPO (2021b).

Figure 11.1 “Economies of provenance” of counterfeit goods

Furthermore, an analysis of imports into the European Union highlights the significance of intermediary transit points (OECD/EUIPO, 2016). These points include well-established international trade hubs like Hong Kong (China), and Singapore. Additionally, as discussed above, transit points can be economies with weak governance or a notable presence of organized criminal or even terrorist activity. The landscape of transit points can change significantly from year to year as traffickers adapt to exploit new governance gaps. This highlights the ability of counterfeiters and criminal networks to swiftly identify vulnerabilities and leverage them for their illicit activities. As noted in the previous section, many of these transit points also host large free trade zones, which may inadvertently facilitate the movement of counterfeit goods.

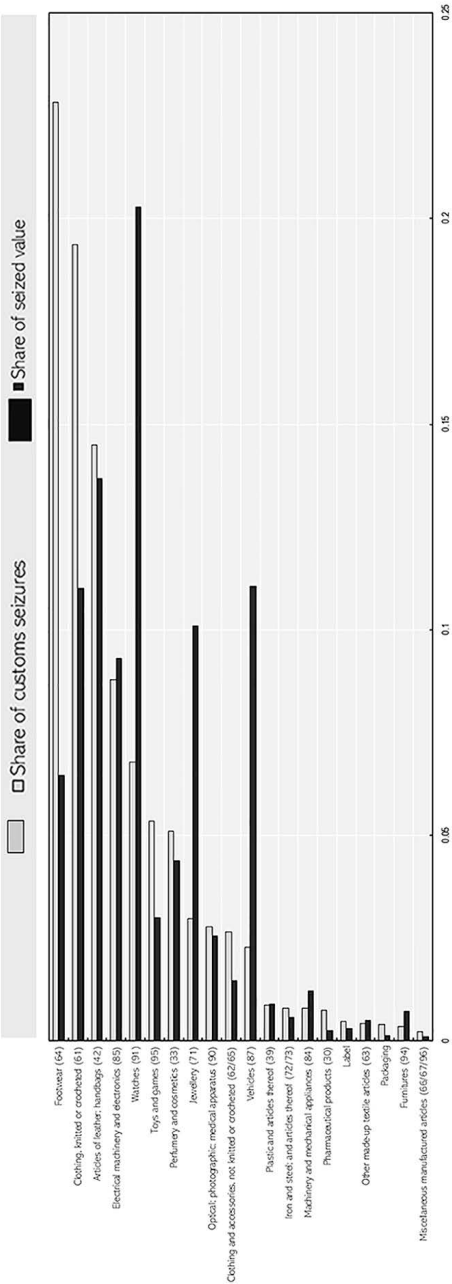
5.4 Products Impacted

Every IP-protected product becomes a target for criminals, and the quantitative analysis conducted by the OECD and EUIPO reveals that intellectual property infringement occurs in various sectors. Between 2017 and 2019, customs detected IP-infringing articles in 83 out of the 96 product categories used in international trade statistics (Harmonized System, or HS chapters), indicating that counterfeiters target almost any type of product.

Figure 11.2 presents the main product categories (HS chapters) ranked according to the frequency of seizures of counterfeits. As in the case of “economies of provenance”, two indicators are used: the share of seizures in a given HS chapter in the total number of seizures (irrespective of their size or value), or the share of the total value of seizures in a given HS chapter in the total value of global seizures.

IP infringements encompass a broad scope and affect diverse categories of goods, including common consumer products like clothing, food, cosmetics, and footwear. They also extend to specialized consumer products such as pharmaceuticals and technical devices, business-to-business products like spare parts and batteries, and luxury items such as fashion apparel and deluxe watches. Counterfeiting also affects goods that do not seem to be initially associated with this risk, including fresh strawberries, breathing apparatus, pizza ovens, artificial grass, guitars, fish food, and construction materials. This expanding industry scope illustrates the aggressive strategies employed by criminals that breach IP rights as they seek profit opportunities.

As discussed earlier, criminals employ techniques such as shipping genuine goods separately from IP-infringing packaging to reduce the risk of detection. This practice is confirmed by the descriptive analysis of the seizures database that reveals a substantial number of seized counterfeit labels. In 2019 alone, the unified dataset included nearly 6,000 customs seizures of counterfeit labels, representing a 20% increase since 2016 (OECD/EUIPO, 2021b).



Note: The number in brackets indicates the HS (Harmonized System) code of a given product category polluted by IP-infringing goods.
Source: OECD/EUIPO, 2021b.

Figure 11.2 Product categories impacted by counterfeiting

Nevertheless, as shown in Figure 11.2, there are indications that interceptions of fake goods are not evenly distributed across all product categories. Some sectors, particularly those that heavily rely on patents – such as pharmaceuticals, electrical machinery, and electronic equipment – are more frequently targeted by counterfeiters. Taking electronic equipment as an example, detailed data analysis finds that smartphone batteries, chargers, memory cards, magnetic stripe cards, solid state drives and music players are also increasingly falling prey to counterfeiters. On average, 6.5% of global trade in information and communication technology (ICT) goods is in counterfeit products, according to analysis of 2013 customs data (OECD, 2017). Alarmingly, many IP-infringing goods in these categories pose genuine threats to consumer health, safety, and the environment.

Counterfeit medicines are of particular concern, as in each instance of IP infringement the associated health risks can be particularly severe. The trade in fake pharmaceuticals has been extensively analyzed in a dedicated report by the OECD and EUIPO (OECD/EUIPO, 2020). This report reveals that both common medicines like painkillers and more complex treatments in areas such as oncology and cardiovascular diseases are subject to counterfeiting.

Apart from fake medicines there are many other product categories where IP-infringing goods can also pose risks to consumer health and safety, including for example fake foodstuffs, toys, cosmetics, and chemicals. Consuming counterfeit food products can have detrimental effects, while counterfeit toys may lack safety measures and pose hazards to children. Similarly, counterfeit cosmetics can contain harmful substances, and counterfeit chemicals like fertilizers or pesticides can lead to environmental issues (OECD/EUIPO, 2022).

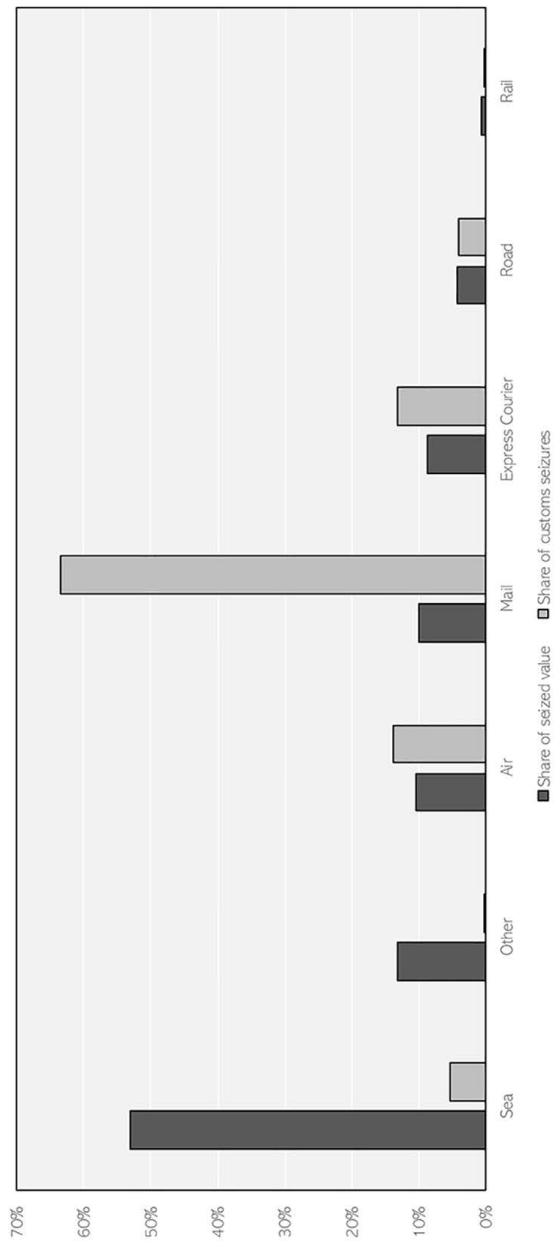
To reiterate the risks associated with IP infringement go beyond economic implications and extend to the well-being and safety of consumers. Protecting intellectual property becomes crucial not only for businesses but also for safeguarding public health and preserving the integrity of various industries.

5.5 Conveyance Methods

Illicit trade networks exploit various modes of transportation and utilize every means available to ship fake goods.

Figure 11.3 presents the conveyance methods abused by criminals to smuggle IP-infringing goods. As with the previous illustrations there are two indicators that can be used: the share of seizures in each category of transport (irrespective of their size and value), or the share of the total value of seizures in each category of transport.

Data analysis has revealed that in terms of frequency of seizures small parcels, particularly those transported through postal services, are the most abused method. Between 2017 and 2019, postal shipments accounted for 64%



Source: OECD/EUIPO (2021b).

Figure 11.3 Conveyance methods of IP infringing goods

of global seizures, with an additional 13% involving express courier services. Consequently, small parcels, including both postal and express services, have significantly dominated the trade in IP infringing goods, constituting 77% of global seizures.

Inspecting small parcels poses a substantial challenge for customs officers, as it requires significant resources. Effective inspection of these parcels is particularly difficult and costly for most countries. This presents a significant enforcement hurdle when dealing with the abuse of small parcels.

In terms of value, IP-infringing goods transported by container ships hold the majority share, accounting for over half of the global value of counterfeit seizures in 2019. However, the observed trends indicate a continued increase in the use of small parcels. Additionally, on average, seized shipments tend to contain fewer IP-infringing goods, with shipments containing fewer than 10 items comprising approximately 85% of the total number of seizures on average. This is a notable shift compared to the 2011–2013 period when such small shipments accounted for only 43% of the total. Consequently, small parcels, often carrying only a few items, remain the preferred method for transporting counterfeit and pirated products (OECD/EUIPO, 2021b; see Figure 11.3).

These trends highlight why detecting IP-infringing goods has become more difficult and costly. As criminals adapt their strategies and exploit the limitations of inspection procedures, it becomes increasingly challenging for authorities to detect and seize counterfeit and pirated goods transported through these means.

5.6 Existing Trends and Expectations

Illicit trade in IP-infringing goods is a dynamically-changing risk, as criminals swiftly adapt their methods to the current conditions. The COVID-19 pandemic has further exacerbated the issue of illicit trade in IP-infringing goods. The crisis correlated with a surge in the misuse of online platforms, as consumers in lockdown turned to online markets to meet their needs. This has led to a massive increase in the availability of IP-infringing goods across a wide range of goods, including medicines, personal protective equipment (PPE), watches, consumer goods, and electrical machinery. Criminal networks swiftly adapted their strategies to exploit the changing circumstances, ensuring a continuous supply of illicit goods during lockdowns. These organized criminal groups demonstrated their ability to anticipate disruptions in transport routes and adjusted their operations accordingly. The staggered impact of the pandemic across different regions also allowed them to learn from early affected areas, such as Far East Asia.

Furthermore, there has been a notable shift towards increased misuse of the online environment. With lockdowns implemented in many countries,

online platforms have experienced intensified usage as digitization rates soared. Criminals have taken advantage of this trend, resulting in a significant growth in the availability of fake products across various online sites and platforms, including social media. The surge in e-commerce activity, with the United States seeing a 146% increase compared to the previous year, has further facilitated the supply of counterfeit goods online. This has also led to the emergence of new online marketplaces and platforms during the pandemic. Criminals have expanded their presence on websites related to mechanical and electrical engineering, and metalworking industries, and have begun misusing online communication channels like WhatsApp and Facebook Messenger (OECD, 2021, OECD/EUIPO, 2021c and UNICRI, 2020).

Finally, the impact of COVID-19 on IP infringement concerned a wide of range of health goods. With lockdown restrictions, e-commerce has become a prominent platform for the sale of fake and substandard medicines, protective equipment, sanitizers etc. Fake medical products, particularly those related to COVID-19, are often purchased online and shipped in small parcels via air cargo.

6. CONCLUSIONS

Illicit trade in IP infringing goods, including patent infringing goods, poses a significant, global risk. The use of IP baskets by modern companies results in illegal products infringing on many different IPs at once, including patents. The volumes are significant, with an estimated value of USD 464 billion in international trade in counterfeit and pirated products in 2019 alone. This accounts for up to 2.5% of global trade, indicating the extensive reach of counterfeiting across various industries. IP infringement, including patent infringement, impacts all types of products, including common consumer goods, business-to-business products, IT goods, luxury items, and even critical items such as contact lenses, pharmaceuticals, and baby formula. These illicit goods continue to be supplied through multiple channels, leading to high levels of consumer deception, particularly in these categories.

Illicit trade in IP infringing goods is of concern in today's global markets. Several factors, including the economic downturn, the expansion of online marketplaces, and the increase in direct sales to consumers, pose significant challenges to existing anti-IP infringement measures. These developments facilitate counterfeiters' ability to target consumers directly.

The growth of e-commerce has created new opportunities for criminals, making it easier to list and sell their illicit, IP-infringing products to unsuspecting consumers. The substantial volume of online transactions and the rapid rate of online sales render the effective monitoring and detection of counterfeit listings particularly challenging. The internet, with its inherent

anonymity and global reach, significantly facilitates the sale and distribution of counterfeit products. IP infringers are a threat to brand value, undermine consumer trust, affect the country's gross domestic product and pose a threat to health and safety.

Current measures to combat IP infringement, encompassing both organizational and technological strategies, have been inadequate in curtailing the expansion of the illicit market. The scarcity of practical guidance in existing literature exacerbates this challenge, leaving a gap in effective managerial strategies for combating these infringements. This shortfall in research hinders the development of informed managerial responses to the counterfeit dilemma, underlining a critical need for comprehensive strategies that go beyond mere acknowledgment of the counterfeit market's scale (Staaake et al., 2009).

The reliance on estimates to gauge the size of markets for IP infringing goods, while highlighting the issue's significance, falls short of providing the necessary insights for crafting detailed countermeasures within the private sector. The lack of precise data on the financial impact of counterfeits, including effects on brand loyalty and quality perception, leads companies to make decisions based on intuition rather than rigorous investment analysis. This absence of solid data and analytical methods for estimating revenue loss and damage to goodwill makes prioritizing investments in specific markets or countermeasures challenging, thereby hindering the development of effective monitoring and prevention strategies.

Understanding the operations and strategies of bad actors that pollute supply chains with IP-infringing goods is essential for developing effective anti-counterfeiting measures. A simplistic view of criminals as merely seeking quick profits overlooks the complexity of their operations, which include sophisticated production, shipping, and selling tactics. This nuanced understanding reveals vulnerabilities that can be exploited by brand owners. However, many companies fail to adequately consider these factors, often simplifying IP infringement to opportunistic acts rather than strategic, entrepreneurial endeavors (Hakim, 2021).

Furthermore, companies operating globally encounter significant barriers to enforcing IP rights, including patents, particularly when dealing with infringements originating from abroad. Despite the logical recourse to law enforcement in for combating criminal activities such as IP infringements, the effectiveness of such measures is limited when infringers operate outside a given jurisdiction.

Industries that utilize various forms of IP employ strategies to enhance enforcement effectiveness. For instance, incorporating copyrighted materials into products and utilizing trademarks to engage customs authorities demonstrates a proactive approach to leveraging existing legal frameworks. Nonetheless, the key task for developing comprehensive strategies to counteract

IP infringement threats lies with the public sector, underscoring the need for action in supporting the private sector's enforcement efforts.

The huge scale of the threat, and its widespread scale imply that illicit trade in IP infringing goods represents a pervasive threat that thrives due to insufficient enforcement, inadequate resources, weak governance structures, and a lack of information sharing. The COVID-19 pandemic has further highlighted the challenges in combating IP infringement, including patent infringement: criminals have exploited the online environment and are adapting their strategies to continue their illicit trade. It is essential to strengthen international cooperation, enhance enforcement capabilities, and invest in robust governance frameworks to effectively address the issue of counterfeiting and safeguard public health and safety.

The issue of IP infringement is a sensitive one that needs to be addressed with utmost importance, and it is imperative to adopt a comprehensive approach and put in place stricter norms to deal with fakes. Governance plays a critical role in either amplifying or mitigating the prevalence of trade in counterfeit goods. Governance factors, in addition to their direct effect, also multiply the influence of other elements (e.g. presence of FTZs, quality of logistic facilities, or trade facilitation policies) on the share of counterfeits in exports. Therefore, good governance could counterbalance the effects of logistics, trade measures, or FTZs on illicit trade in IP-infringing goods.

A holistic approach necessitates the consideration of all facets of the counterfeiting ecosystem, encompassing prevention, detection, enforcement, education, and collaboration. This approach requires a synergistic combination of legal measures, enhanced enforcement efforts, public awareness campaigns, technological advancements, and international cooperation. By fostering collaboration among governments, law enforcement agencies, intellectual property owners, online platforms, and consumers, a unified front against intellectual property infringement can be established. Such collaborative efforts can result in more effective detection and enforcement actions, improved information sharing, and the development of best practices that can be implemented on a global scale.

To reiterate, the complexity of global supply chains makes them susceptible to penetration by criminal actors. Factors such as corruption levels, intellectual property protection, governance quality, logistics facilities, and trade facilitation policies all influence the propensity for trade in IP infringing goods. Enhancing governance and institutional quality, improving enforcement of laws and regulations, and strengthening intellectual property protection are vital steps in reducing the volume of exports in fakes and mitigating the negative impacts of trade in IP infringing goods on economies.

NOTES

1. See for example “Stamping it out”, *The Economist*, April 23, 2016. <https://www.economist.com/international/2016/04/23/stamping-it-out>
2. See for example reports published by the Transnational Alliance to Combat Illicit Trade (TRACIT) at <https://www.tracit.org/>
3. The term “n.e.c.” stands for “not elsewhere classified”.

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